

SSH commented speedrun

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NOT NULL
Makers

SSH Components

Client `ssh`



Server `sshd`



Connect

```
ssh user@destination
```

Like e-mail 🙄

Connect

```
ssh root@2.28.229.111
```

```
ssh abel@2a01:4f8:1c18:8773::1
```

```
ssh bob@training.notnullmakers.com
```



nnm-101-<user>

abel bob carmen dido elias fred gustav
hank ian jules karel luna mia nina
oleg paul quentin richard simon tomas
ursula vit walter xena yale zara adam
bert cameron david eva felix gabriel
hunter ivan jana kevin leo mark nadia
omar pablo qasim rose saul tara uma
vivian wren xavier yael zelda



nnm-101-<user>

Connect 🤨

The authenticity of host
'training.notnullmakers.com (2.28.229.111)' can't
be established.
ED25519 key fingerprint is
SHA256:QX9HDKKeDpSVQ9PFKZ6s5TvvUAkvCIBHlRw0A38ePoH8.
This key is not known by any other names.
Are you sure you want to continue connecting
(yes/no/[fingerprint])?

Keys

```
ssh-keygen -t ed25519 -C "training"
```

-> /home/user/.ssh/id_ed25519

-> /home/user/.ssh/id_ed25519.pub

More Keys

```
ssh-keygen -t ed25519 -C "training" \  
-f training_key -N ''
```

Authorized Keys

`~/.ssh/authorized_keys`

`ssh-copy-id -i <pubkey> user@host`

```
ssh-copy-id -i .ssh/id_ed25519 \  
    <user>@training.notnullmakers.com
```

Authorized Keys

```
cat ~/.ssh/id_ed25519.pub | \
ssh user@host "mkdir -p .ssh && \
cat - >> .ssh/authorized_keys"
```

Known Hosts

~/.ssh/known_hosts

```
|1|+cerzND2aF0gYWkP2/HT41sQ0Lk=|  
aC0FWYM9HhjeZTLif7G5I9Q60dU= ssh-ed25519  
AAAAC3NzaC1lZDI1NTE5AAAAID3JIjt4i+tq2If+Qity08pu+  
Qv49Veg0AuVtrIWzh7r  
|1|0Mbjux2v31Jap4mbBZeABYRMo1U=|  
fEcByJzGBqFc1YX9eVEHdhQ2CHY= ssh-rsa  
AAAAB3NzaC1yc2EAAAADAQABAAQgQC1TLqUZs0Nda1wXuP98  
nFZSuPODZhywuC6dz0Vqgd9rrJk8x0+gDgpa8aWBfWP0oxMsA  
Zw1mwDwj905L2t0013LTk...
```

Host Keys

```
/etc/ssh/ssh_host_ecdsa_key  
/etc/ssh/ssh_host_ecdsa_key.pub  
/etc/ssh/ssh_host_ed25519_key  
/etc/ssh/ssh_host_ed25519_key.pub  
/etc/ssh/ssh_host_rsa_key  
/etc/ssh/ssh_host_rsa_key.pub
```

SSH Config

`~/.ssh/config` or `/etc/ssh/ssh_config`

```
Host europe
```

```
HostName 2.28.229.111
```

```
User <user>
```

```
IdentityFile %d/.ssh/id_ed25519
```

```
IdentitiesOnly yes
```

Connect Revisited

~~ssh root@2.28.229.111~~

~~ssh abel@2a01:4f8:1c18:8773::1~~

~~ssh bob@training.notnullmakers.com~~

ssh europe



SSH-Agent

```
ssh-add ~/.ssh/id_ed25519
```

```
ssh-add -d ~/.ssh/id_ed25519
```

Copy File To Remote Destination

```
scp <file> europe:~/<file>
```

Copy Files To Remote Destination

```
scp <file1> <file2> europe:~/
```

```
scp -pr <dir> europe:~/
```

SFTP

Interactive or batched

Rsync

Saving data transfers

Piping Over SSH

cat, dd, tar

Training Setup



europe



west-coast

ProxyJump - Intermediary Host

```
ssh -J europe <user>@west-coast
```

-> west-coast in /etc/hosts

Tunneling over SOCKS 🧦

```
ssh -D 9999 -J europe \  
  <user>@west-coast
```

Other Tunneling Options

- Local Port Forwarding (-L)
- Remote Port Forwarding (-R)

Tunneling RDP

- Tunnel Remote Desktop Protocol through SSH

```
ssh -L 5000:win-host:3389 \  
-J jump-host <user>@box-that-sees
```

Tunneling CIFS/Samba

```
ssh -L 5445:win-host:445 \  
-J jump-host <user>@box-that-sees
```

Only SFTP Server

In `/etc/ssh/sshd_config`

```
Subsystem sftp internal-sftp
```

```
Match Group sftponly
```

```
    ChrootDirectory /uploads
```

```
    ForceCommand internal-sftp
```

```
    PermitTunnel no
```

```
    X11Forwarding no
```

```
    AllowTcpForwarding no
```

Only SFTP Server Continued

```
groupadd sftponly
```

```
useradd -d /uploads -g sftponly \  
-s /sbin/nologin -M upload
```

```
mkdir -p /uploads/new
```

```
chown upload:sftponly /uploads/new
```

```
passwd upload
```

```
systemctl restart sshd
```

Only SFTP Server Continued

```
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```

```
useradd -d /uploads -g sftponly \  
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mkdir -p /uploads/new
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chown upload:sftponly /uploads/new
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systemctl restart sshd
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Forwarding and Reverse Proxying

- nftables - Network Address/ Port Translation
- nginx - Reverse proxy

Thank You 🎉

For later questions, feedback:
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